OpenShift Resource Common Metadata

Summary

In OpenShift, labels and annotations provide a means for attaching metadata to Kubernetes objects. While they are similar in that they are key/value pairs, their uses differ. The following provides a description of labels and annotations, as well as the required labels and annotations for given Kubernetes objects /resources.

Reference Documentation

- Monitoring OpenShift | Pod annotations
- OpenShift HAProxy Ingress Annotations - Cloud Services - FCB Confluence Prod

Descriptions

Labels and Selectors

Labels are key/value pairs that are attached to objects such as Pods. They enable users to map their own organizational structures onto system objects in a loosely coupled fashion, without requiring clients to store these mappings. Labels can be used to organize and to select subsets of objects. Labels can be attached to objects at creation time and subsequently added and modified at any time. Each Key must be unique for a given object.

Unlike names and UIDs, labels do not provide uniqueness. In general, we expect many objects to carry the same label(s). Via a label selector, the client/user can identify a set of objects. The label selector is the core grouping primitive in Kubernetes

Annotations

Annotations are means of attaching arbitrary non-identifying metadata to Kubernetes objects, where client tools and libraries can retrieve this metadata. In contrast to labels, annotations are not used to identify and select objects. The metadata in an annotation can be small or large, structured or unstructured, and can include characters not permitted by labels. It is possible to use labels as well as annotations in the metadata of the same object.

Label Guidelines

Namespaces

The following labels are **required** on all namespaces created on the OpenShift platform.

- kubernetes.io/metadata.name: <team input>-<environment>-<zone>
- fcbint.net/network: <network-zone>
- fcbint.net/app-tier: <correlate to CMDB Application>
- fcbint.net/app-id: <CMDB Application ID>

All Other Objects/Resources

The following labels are **required** on all other objects and resources created on the OpenShift platform. This includes ingress, services, all workload types, configmaps and secrets.

- app.kubernetes.io/name:
- app.kubernetes.io/instance:
- app.kubernetes.io/component:
- fcbint.net/app-id: <CMDB Application ID>

Recommended

The following labels are **recommended** on all objects and resources created on the OpenShift platform.

- app.kubernetes.io/version: <semantic version>

Annotation Guidelines

Namespaces

The following annotations are **required** on all namespaces created on the OpenShift platform.

- fcbint.net/contact: <email distribution list>
- fcbint.net/ritm: <RITM000000>
- collectord.io/index: <splunk index for log forwarding>
- openshift.io/requester: <requestor name>

Ingress

The following annotations are **recommended**, as applicable, on Ingress resources created on the OpenShift platform.

- See: OpenShift HAProxy Ingress Annotations - Cloud Services - FCB Confluence Prod

Certificates

The following annotations are **required** on all Keyfactor-issued certificate objects created on the OpenShift platform. This is in compliance with the Keyfactor Certificate CMDB Tracking (Link: TODO).

- metadata.command-issuer.keyfactor.com/Certificate-Support-Group-Contact: (team email address)
- metadata.command-issuer.keyfactor.com/Environment:
  - Non-Prod-DEV
  - Non-Prod-QA
  - Non-Prod-UAT
  - Prod-Corp
- metadata.command-issuer.keyfactor.com/Description: (Free form - describe your service)
- metadata.command-issuer.keyfactor.com/Organization: FCB
- metadata.command-issuer.keyfactor.com/Application-ID: <CMDB Application ID>
- metadata.command-issuer.keyfactor.com/Platform: OPENSHIFT

Workloads

The following annotations are **recommended**, as applicable, on workload resources (Deployments, Statefulsets, Jobs, CronJobs, etc).

- See the following for log control overrides: Monitoring OpenShift | Pod annotations